

Impact of Generative AI on Student Academic Performance and Learning Outcomes

Executive Summary

This report analyzes the impact of Generative AI usage on students' academic performance, learning behavior, skill retention, and well-being. The analysis was conducted using Python for data exploration and statistical testing, and Power BI for dashboard development. The findings highlight that while AI is widely adopted across academic disciplines, excessive usage does not necessarily lead to better academic outcomes.

1. Project Objective

The objective of this study is to understand how students use Generative AI tools and evaluate their influence on GPA, study habits, skill retention, anxiety levels, and burnout risk. The project aims to identify whether AI acts as a learning enhancer or creates dependency-related challenges.

2. Dataset Overview

The dataset contains student demographics, academic performance indicators, AI usage metrics, learning behavior measures, and well-being indicators. Key variables include Weekly_GenAI_Hours, Pre and Post Semester GPA, Skill Retention Score, AI Dependency, Burnout Risk Level, and Primary Use Case.

3. Analysis Performed

- Data Cleaning and Validation
- Exploratory Data Analysis (EDA)
- Distribution Analysis of AI Usage and GPA
- Major-wise AI Adoption Analysis
- Primary Use Case Analysis
- Burnout Risk Assessment
- AI Dependency vs Skill Retention Analysis
- Hypothesis Testing using Independent T-Test
- Interactive Dashboard Development in Power BI

4. Key Dashboard Insights

AI Adoption by Academic Major:

STEM students recorded the highest average weekly AI usage (approximately 10.49 hours), followed by Business students. This indicates stronger adoption of AI tools in technical and analytical disciplines.

Primary Use Cases of AI:

Students primarily use AI for assignments, research support, coding assistance, and exam preparation. The findings suggest that AI is increasingly becoming a productivity and learning support tool.

Academic Performance Trends:

GPA improvement varied across usage categories. Moderate AI usage appeared more beneficial than excessive usage, suggesting that balanced adoption may provide the greatest academic value.

Burnout and Well-Being:

Burnout risk distribution highlighted that a considerable proportion of students experience moderate to high burnout risk, emphasizing the importance of healthy study practices and responsible AI usage.

5. Statistical Analysis

An Independent T-Test was performed to compare Post Semester GPA between Low AI Users and Heavy AI Users.

Results:

- T-Statistic = 3.485
- P-Value = 0.00049
- Low Users Mean GPA = 3.342
- Heavy Users Mean GPA = 3.320

Interpretation:

Since the p-value is less than 0.05, the difference between the two groups is statistically significant. However, the actual GPA difference is very small (approximately 0.02 GPA points). This indicates that while the relationship exists statistically, the practical impact on academic performance is limited.

6. Conclusions

The analysis suggests that Generative AI has become an important academic support tool for students. However, excessive usage does not automatically translate into better academic outcomes. Students who rely heavily on AI do not significantly outperform low users in terms of GPA. The findings indicate that balanced and purposeful AI usage is more beneficial than overdependence.

7. Recommendations

- Promote responsible and balanced AI usage among students.
- Provide AI literacy and prompt engineering training.
- Encourage critical thinking alongside AI-assisted learning.
- Monitor skill retention rather than focusing solely on GPA outcomes.
- Develop institutional guidelines for ethical and effective AI use.
- Track student burnout and anxiety levels to support well-being.

Prepared as a Data Analytics Project using Python, Statistical Analysis, and Power BI.